Unit Test Report

Blood Bank Management System (BBMS) — Team 9, Section C

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Reviewed by	Dheepak Seetharam (PES1UG24CS152, QA Lead)
Branch	`feature/sprint-2-integration`
Commit	`18e1374`
Related Issues	#74, #76
References	SRS v1.1, SAD v1.0, STP v1.0, RTM

1. Purpose & Scope

This report records the results of the automated unit and integration test suites for BBMS at the end of Sprint 2. It covers the Express backend (`backend/src/**`) and the React frontend (`frontend/src/**`).

In scope: business logic, middleware, repositories, route handlers (tested at the HTTP boundary), report exporters, frontend components, pages, services and utilities.

Out of scope: manual cross-service workflows (see the System Validation Report), load and stress testing, and penetration testing.

2. Test Environment & Tools

Item	Value
Backend runtime	Node.js 20+ (verified with 24.15.0), Express 4
Backend test tools	Jest 29.7.0, Supertest, ESLint
Frontend test tools	Vitest 1.6.1, `@vitest/coverage-v8` 1.6.1
Database	PostgreSQL 17 on `localhost:5433`, database `blood_bank`
Test data	Deterministic: `npm run db:reset` then `npm run db:seed`
Coverage engines	Istanbul (backend), V8 (frontend)
Report artifacts	`docs/reports/unit-test-report.html` (Jest HTML), `docs/reports/code-coverage/`, `docs/reports/frontend-coverage/`

Suites that need a live PostgreSQL use `describeIfDb` and skip automatically when `DATABASE_URL` is unset, so CI without a database still passes.

3. Test Strategy

Layer	Tooling	What is verified
Backend unit	Jest	Pure logic: validators, eligibility engine, stock service, report exporters (CSV/PDF), middleware (RBAC, validation, request ID, session, CSRF)
Backend route / integration	Jest + Supertest	Every API route at the HTTP boundary: status codes, standard error shape, RBAC, CSRF, audit trail, database effects
Frontend unit / component	Vitest	Components (Navbar, forms, dashboards, reports, search), pages, `AuthContext`, `services/api.js`, `utils` (validation, navigation)
Traceability	Jest (`rtmValidation.test.js`)	Every requirement in the RTM maps to at least one test case; runs in CI

Principles followed:

- Tests are deterministic and independent; three consecutive runs gave identical results.
- Every business rule is asserted at the route level so that frontend validation gaps do not leave a rule untested.
- Error paths (401, 403, 404, 409, 423) are tested alongside success paths.

4. Results Summary

Layer	Test suites / files	Tests	Failures
Backend (Jest)	47 suites	461	**0**
Frontend (Vitest)	22 files	52	**0**
Total	**69**	**513**	**0**

Coverage achieved (details in the Code Coverage and Branch Coverage Reports):

Layer	Statements	Branches	Functions	Lines
Backend	**93.15%**	**81.06%**	**97.29%**	**94.52%**
Frontend	**74.54%**	54.12%	36.91%	**74.54%**

5. Backend Tests by Module

Module	Behaviour verified	Representative test files
`auth`	Registration, login with lockout after 5 failed attempts (15 minutes), logout, password reset (one-time 15-minute token), own profile	`auth.register.test.js`, `auth.login.test.js`
`admin`	Create Staff, list/filter users, activate/deactivate, change role, `LAST_ADMIN` and `SELF_ACTION_DENIED` guards, audit-log viewer	`admin.test.js`
`donors`	Donor profile management, eligibility engine, account linking, own donation history	`donors.test.js`
`donations`	Record donation (Passed / Failed screening), unit creation, last-donation update, correction within 24 hours	`donations.record.routes.test.js`
`stock`	Summary by group and status, expiry, low-stock and expiring-soon alerts, discard with mandatory reason	`stock.discard.routes.test.js`

Module	Behaviour verified	Representative test files
`search`	Stock search by blood group, eligible-donor search, error-handler branches	`search.test.js`, `search.repository.test.js`
`requests`	Submit, own requests, pending queue, approve with FEFO, reject, issue, cancel	`requests.routes.test.js`
`reports`	Stock, donation and request reports; CSV and PDF export	`reports.test.js`
`middleware`	Deny-by-default RBAC, input validation, request ID, session, CSRF, error handler	`rbac.test.js`
`audit`	Audit writer; retention job (365 days, daily at 02:00)	See the Jest HTML report
`health`	Health endpoint	See the Jest HTML report
Traceability	RTM completeness check	`rtmValidation.test.js`

The full per-suite list with individual test names is in `docs/reports/unit-test-report.html`.

6. Frontend Tests

Area	Behaviour verified	Representative test files
App shell	`App.jsx` and `main.jsx` render and route correctly	—
Components	Navbar, blood request form, donor registration form, dashboards, reports module, search module	—
Pages	Home, Profile, Stock, Users, Requests queue, Login, Forgot password	—
Context and services	`AuthContext`, centralised `api.js` (CSRF, credentials)	—
Utilities	Form validation rules, role-based navigation	`validation.test.jsx`

Configuration notes:

- Vitest runs with `pool: 'forks'` and `singleFork: true` to avoid worker out-of-memory failures.
- A global `afterEach(cleanup)` in `frontend/vitest.setup.js` unmounts rendered components between tests.

7. Requirements Traceability

`docs/RTM.md` maps all 79 requirements (54 functional, 19 non-functional and 6 business rules) to at least one test case. The RTM validation test (`rtmValidation.test.js`) runs in CI and passes, so a requirement cannot lose its test mapping unnoticed.

8. Defects Found and Fixed This Sprint

ID	Finding	Resolution
D-04	`search.routes.js` had 0% branch coverage: the error handler was never exercised	Tests added that force both the `SearchError` and non-`SearchError` branches; now 100%

Frontend files that started the sprint near 0% coverage (for example `ProfilePage.jsx`, `StockPage.jsx`, `UsersPage.jsx`, `main.jsx`) now have tests; see the Code Coverage Report §4.4 for the before/after figures.

9. Reproducibility

```
# Backend
cd backend && npm install
npm run db:reset && npm run db:seed
npm run test:reports         # Jest + coverage artifacts

# Frontend
cd frontend && npm install
npx vitest run --coverage    # Vitest + coverage artifacts

# Lint
npm run lint
```

10. Known Limitations and Deferred Tests

- **Frontend form validation branches.** Per-field error-path tests for the larger forms (login, registration, donor registration, password reset) are deferred. The same rules are enforced and tested at the API boundary, and the forms are covered by manual system validation.
- **Interactive pages.** Multi-step workflows on the requests queue, stock and users pages are covered at the route level, but their UI interaction branches are not unit tested.
- **Transaction guard clauses.** The `LAST_ADMIN` and `SELF_ACTION_DENIED` guards in `admin.service.js` are tested through the route tests; direct unit tests would need full `withTransaction` mocking.
- **No automated end-to-end suite.** Cross-service workflows are validated manually (see the System Validation Report).

The Branch Coverage Report lists these gaps with justification and a Sprint 3 improvement plan.

11. Conclusion

All 513 automated tests (461 backend, 52 frontend) pass with zero failures. Backend statement coverage is 93.15% and frontend statement coverage is 74.54%, both above the project target of 70%. All business-critical modules are covered at 90% or more. The suites are deterministic and run in CI.

12. Sign-off

Role	Name	SRN	Status
Prepared by	Harsimran Kaur (Product Owner)	PES1UG24CS187	Pending
Reviewed by	Dheepak Seetharam (QA Lead)	PES1UG24CS152	Pending
Approved by	Chukkapalli Rohan (Test Engineer)	PES1UG24CS135	Pending

End of report.